

Training-Free Adaptive Sampling Scheduler for Discrete-Time Consistency Distillation

One scheduler, no retraining — drop it into LCM, PCM, TCD or TDD.

Qi Wang · Jinjia Zhou

Graduate School of Science and Engineering · Hosei University · Tokyo, Japan



Scan: full summary

1 Why few-step sampling still breaks

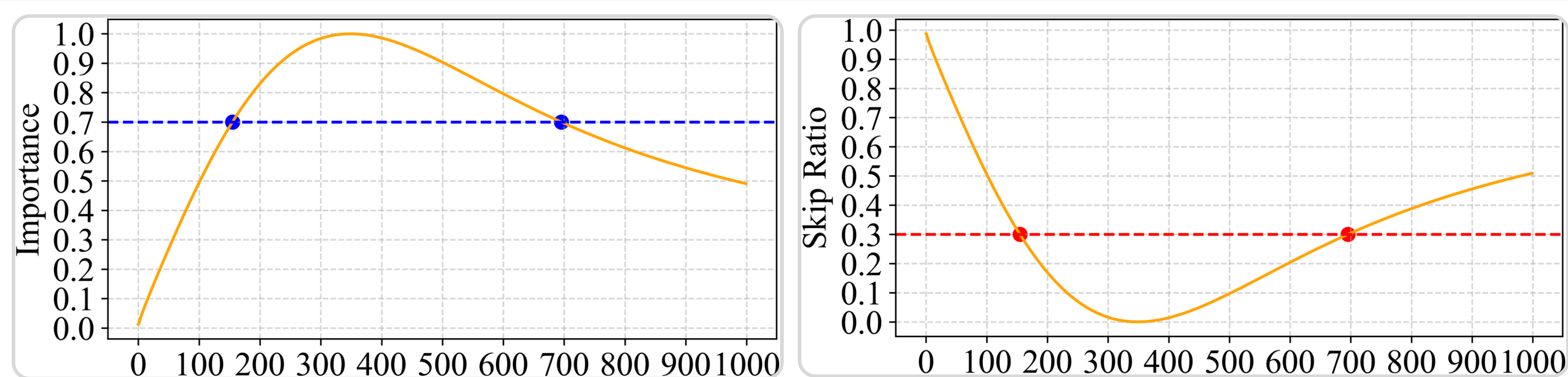
- **Scheduler rigidity.** Every discrete-time method ships a schedule welded to its own framework and step budget; deterministic- (LCM, PCM) and stochastic-target (TCD, TDD) strategies alike degrade as soon as the scheduler or the step count changes.
- **CFG compounding.** Inference guidance multiplies with the inner distillation scale, pushing pixels into saturation and colour drift.

2 Timesteps are not equally informative

In the forward process the image stays legible to $t \approx 200$ and is indistinguishable from noise past $t \approx 800$; nearly all structural change is packed in between. We quantify this with an **Importance** I_t — the rate of change of log-SNR, inverted and normalised to $[0, 1]$. A large I_t marks a timestep where SNR moves fast; the complementary **skip rate** R_t says how much of that step can safely be skipped.

IMPORTANCE & SKIP RATE (EQ. 8) | $\text{SNR}_t = \bar{\alpha}_t / (1 - \bar{\alpha}_t)$, $\varepsilon = 10^{-8}$

$$I_t = \frac{|\nabla_t \ln(\text{SNR}_t + \varepsilon)|^{-1}}{\max_{0 \leq j < T} |\nabla_j \ln(\text{SNR}_j + \varepsilon)|^{-1}}, \quad R_t = 1 - I_t + \varepsilon$$



(a) I_t peaks near $t \approx 350$; the dashed line is $\theta = 0.7$, whose two crossings bracket the band worth sampling densely. (b) The skip rate $R_t = 1 - I_t$ mirrors it.

3 Adaptive schedule & γ -I sampler ★ KEY

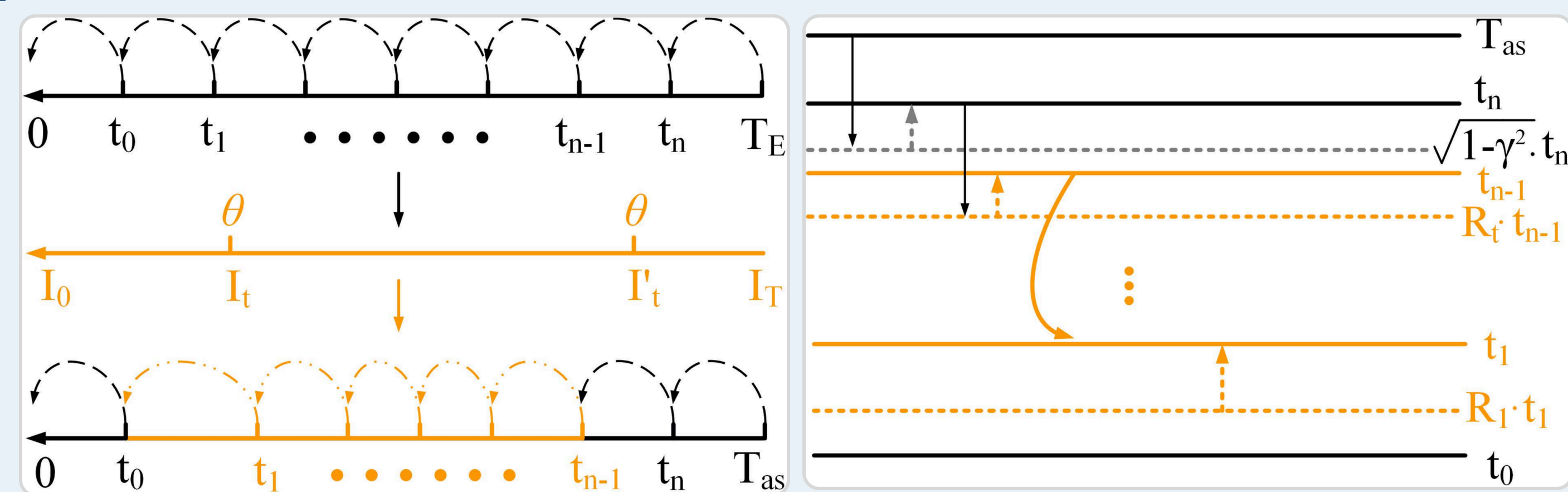
The schedule: keep uniform-grid timesteps T_E where change is slow, substitute high-Importance candidates T_I where it is fast. The **step count never changes**, only its spacing. **The sampler:** instead of CTM's fixed γ , each step scales its noise re-injection by the local Importance, widening exploration where the trajectory turns fastest.

ADAPTIVE SCHEDULE (EQ. 9), $\theta = 0.7$

$$T_{AS} = \{t_i \in T_I \mid I_{t_i} > \theta\} \cup \{t_i \in T_E \mid I_{t_i} \leq \theta\}$$

UNIFORM STEP (EQ. 10) / HIGH-IMPORTANCE STEP (EQ. 11)

$$x_{t_{n+1}} \xrightarrow{f_\theta} \sqrt{(1-\gamma)^2} \cdot t_n \rightarrow x_{t_n}, \quad x_{t_n} \xrightarrow{f_\theta} I_t \cdot t_{n-1} \rightarrow x_{t_{n-1}}$$



(a) Dense sampling in high-Importance regions, sparse elsewhere, at a preserved budget. (b) Importance-modulated forward and backward transitions lift sampling quality precisely where the SNR changes most rapidly.

4 Smoothing & colour balancing

To undo CFG compounding we squash outliers with $\tanh$, then re-centre channel and global means so values return to its **linear region** instead of pinning at ± 1 — no prior deformation, and more information survives under strong guidance.

SOFT CLAMP & RE-CENTRING (EQ. 12–13), $\alpha = \beta = 0.5$

$$x_0 = \frac{e^{x_0} - e^{-x_0}}{e^{x_0} + e^{-x_0}}, \quad x_c \leftarrow x_c - \alpha \text{mean}(x_c), \quad x_0 \leftarrow x_0 - \beta \text{mean}(x_0)$$

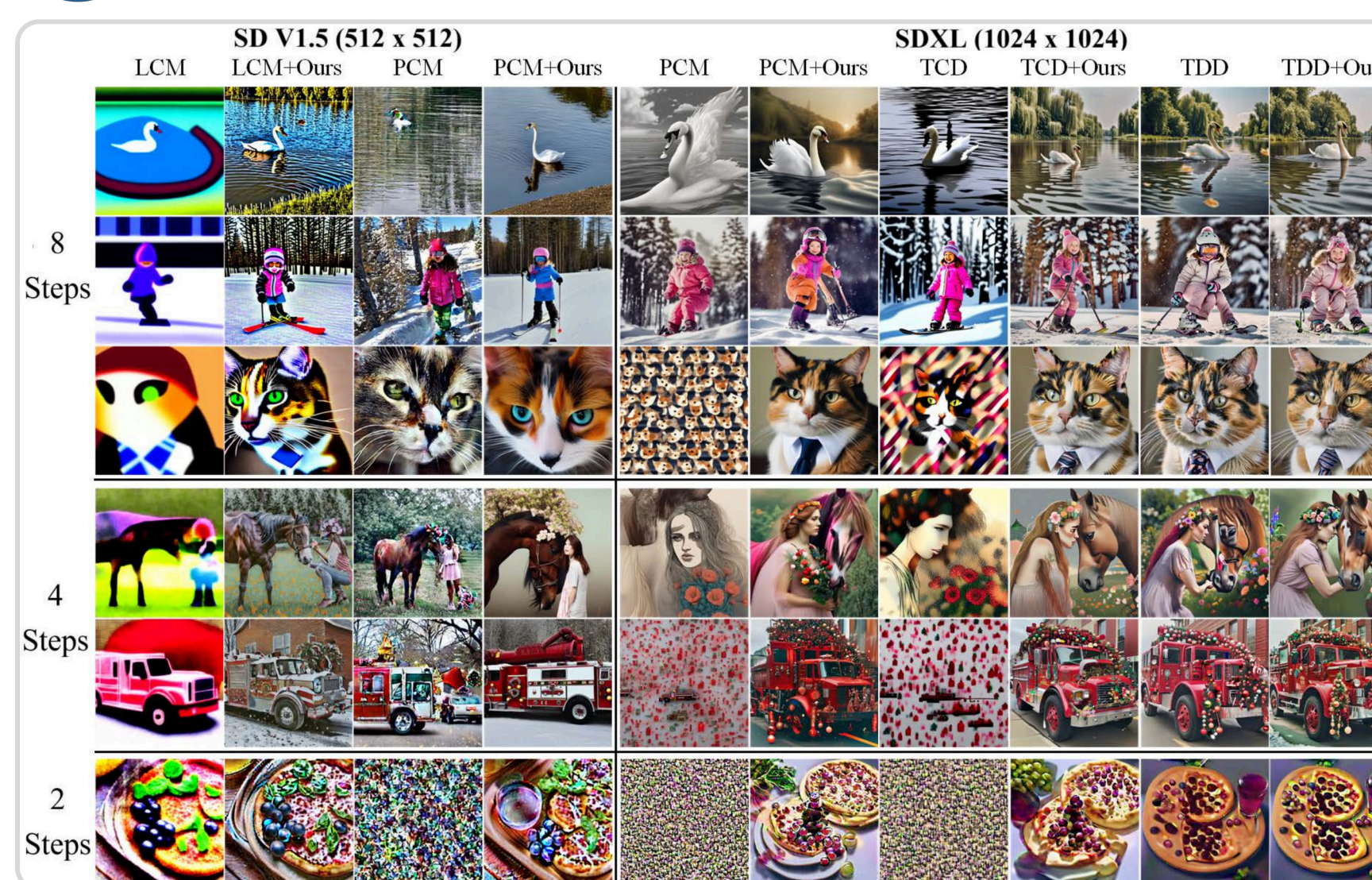
5 Quantitative results ★ Headline

MS-COCO 2014 val · Karpathy 30k captions · CFG $\omega = 7.5$ · columns 2 / 4 / 8 steps · Δ MPI = mean gain.

Method	FID ↓			CLIP Score ↑			Inception Score ↑		
	2	4	8	2	4	8	2	4	8
SDXL — 1024×1024									
PCM	372.82	112.65	31.73	18.44	24.18	30.44	1.71	11.67	25.78
PCM + Ours	65.83	29.40	23.21	30.22	30.40	31.52	16.54	24.59	31.80
TCD	363.50	103.66	53.72	18.73	26.05	30.44	1.82	12.47	17.83
TCD + Ours	62.89	28.51	27.44	28.88	31.71	32.06	16.33	28.02	32.17
TDD	58.45	29.80	27.72	29.27	31.12	31.47	17.18	27.02	29.72
TDD + Ours	55.71	27.88	26.60	29.49	31.47	31.74	17.62	29.36	32.88
Δ MPI	203.45	53.44	11.97	7.38	4.08	0.99	9.93	10.27	7.84
SD v1.5 — 512×512									
LCM	86.33	88.20	109.84	28.02	26.48	25.19	14.28	11.73	8.71
LCM + Ours	58.01	30.04	47.67	30.18	30.71	30.79	17.18	28.81	19.84
PCM	424.04	89.99	38.82	18.99	26.51	30.02	1.76	12.49	21.07
PCM + Ours	60.66	23.11	22.47	29.93	30.37	31.03	16.86	28.93	31.65
Δ MPI	195.85	62.52	39.26	6.55	4.05	3.31	9.00	16.76	10.86

Reading the table. All five baselines improve on every metric at every budget, and the gains land where few-step sampling is weakest — at 2 steps FID falls **82–86%** for PCM and TCD. LCM alone gets worse with more steps (86.33 → 109.84 FID): the CFG-overexposure signature, which ours reverses. The pattern holds across both resolutions and all four distilled students, not just on the metric we optimised for.

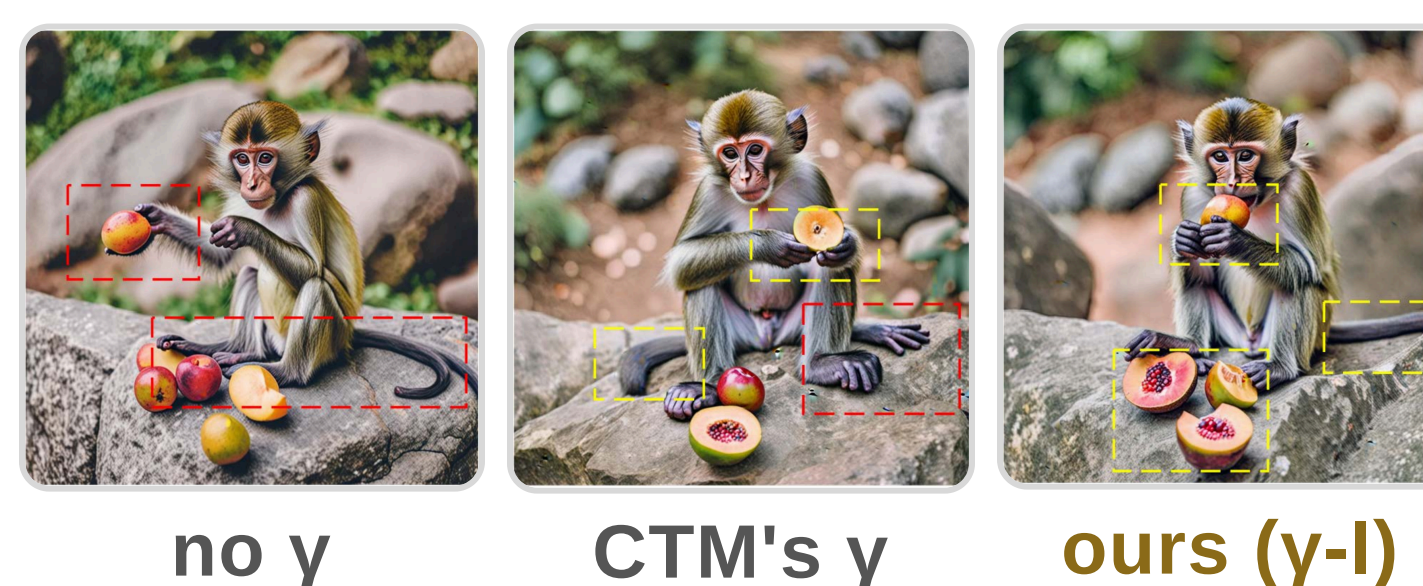
6 Qualitative comparison



What to look for

At **2 and 4 steps** the baselines lose global structure and wash the colour out. Ours keeps the layout coherent and the detail sharper — on **both** backbones. By 8 steps the gap narrows, as expected once the student has budget enough to correct itself.

7 Ablations



The sampler

Scaling noise re-injection by the local Importance, instead of CTM's fixed γ , gives more stable shapes and fewer structural errors at the same step budget. Threshold sweep: best $\theta \in [0.7, 0.8]$.



Smoothing & colour balance

Top row: the clipping used by prior work. **Bottom row:** ours — natural colour, fewer saturated regions and more surviving detail at high CFG.

8 What to take away ★

- **Diagnosis.** The schedule, not the student, is the bottleneck under cross-method or variable-step transfer.
- **Mechanism.** An SNR-derived Importance says where computation is worth spending; selection, the γ -I sampler and a colour pass do the rest.
- **Cost.** No retraining, no extra parameters, negligible runtime overhead — a drop-in scheduler swap on top of an existing distilled student.

In short: training-free, lightweight, drop-in — largest gains at 2–4 steps, consistent across SDXL and SD v1.5, across all four distilled students tested.